

Module 1: Work Order Management

Tab 9: SAP Update

1. Overview

This tab manages the critical process of registering newly installed electrical assets into SEC's SAP system. The workflow is designed to be highly efficient, leveraging photos taken during the Electrical phase and providing a structured interface for data entry and verification. The successful completion of this tab results in obtaining a unique "Equipment Number" for each asset and a single "Functional Position" for the entire Work Order.

2. Workflow

The SAP Update process is as follows:

- 1 **Asset Identification:** The system identifies all electrical assets installed in the project based on the material reconciliation data from the Electrical tab.
- 2 **Asset-Specific Tabs:** For each identified asset, the system generates a dedicated sub-tab within this SAP Update tab.
- 3 **Photo-Assisted Data Entry:** Within each asset's sub-tab, the system displays the relevant GIS photos taken during the Electrical phase. An operator then uses these photos to fill out the Form - Electrical Assets Registration.
- 4 **SAP Registration:** The operator manually enters the data from the completed form into the SEC SAP system.
- 5 **Data Retrieval:** After successful registration in SAP, the operator retrieves the Functional Position and the asset-specific Equipment Number and enters them back into our system.
- 6 **Completion:** The SAP Update tab is marked as complete once every installed asset has a corresponding Equipment Number entered.

3. Main View

The main view of this tab will be a series of sub-tabs, one for each major asset installed (e.g., "Transformer 1," "Cabinet 1," "Cable Link 1").

4. Asset Sub-Tab View

Each asset-specific sub-tab will be divided into two main sections:

4.1. Section 1: Material Information Photos (Read-Only)

- This section will display a gallery of all GIS photos taken for the specific asset during the Electrical phase.
- The photos will serve as the primary source of information for the data entry operator.

4.2. Section 2: Form - Electrical Assets Registration (Input)

- This section will be a dynamic form based on the asset type. The system will use the appropriate worksheet from the 10.02Form-ElectricalAssetsRegistration.xlsx file as a template.
- **Example for a Transformer:**
 - The form will include fields like Substation No, capacity, Primary voltage, Manufacture, Serial No, etc.
 - Many of these fields will be pre-populated with data from the Work Order and material lists.
 - The operator will fill in the remaining fields by visually inspecting the photos (e.g., reading the serial number from a photo of the transformer's nameplate).

5. SAP Data Input

After the registration form is complete, the following fields will become active for the operator to enter the data received from SAP:

- **Field:** Functional Position
- **Data Type:** String
- **Notes:** This is a single value for the entire Work Order and will be the same across all asset sub-tabs.
- **Field:** Equipment Number
- **Data Type:** String
- **Notes:** This is a unique number for the specific asset being registered in the current sub-tab.

6. Completion Logic

- The SAP Update milestone on the main dashboard is marked as **complete** only when every asset listed in the materials_reconciliation table (from the Electrical tab) has a non-null equipment_number associated with it in the sap_asset_data table.